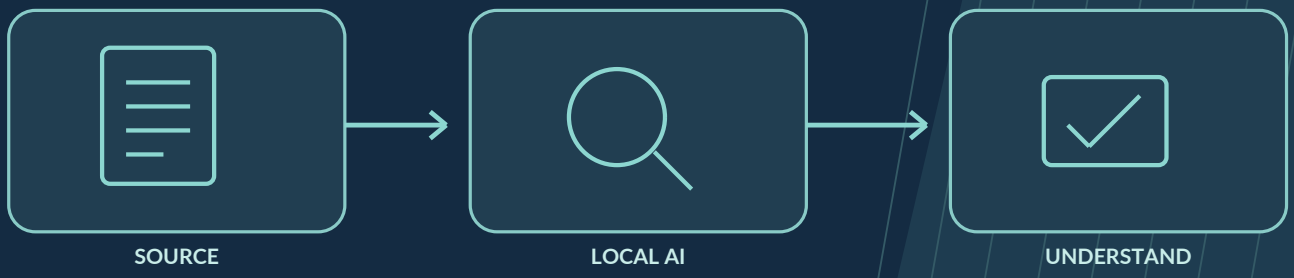
APPBUILDERS PH / LOCAL AI HACKATHON

PirmaBago

Understand what you are signing.

A private, on-device agreement understanding companion for Filipinos. Scan, examine, learn, and prepare questions before making an important commitment - even without internet access.



COMPLETE PRODUCT BLUEPRINT

Product strategy / use cases / requirements / screen inventory / user flows / on-device AI architecture / data model / security / testing / live demo

00 / Executive summary

Private understanding, not automated legal verdicts.

PirmaBago helps people examine important agreements before signing them. A person scans or imports a document; bundled OCR and a quantized language model run on the device to extract terms, explain supported provisions, and create an evidence-linked review. The user can explore documented consequences, test their own understanding, and prepare clarification questions.

> **Core promise**
> Meaningful OCR and language-model inference run on the phone. No account, cloud OCR, or cloud LLM is required for the hackathon workflow. Every material claim must point back to a source passage.

PRODUCT	DECISION
Primary audience	Filipino freelancers, employees, tenants, installment buyers and microbusiness owners
Initial platform	Android ARM64 physical device; Expo / React Native development build
Hackathon focus	Freelance/service agreements and simple installment agreements
Signature features	Evidence-first review, clause-to-consequence explorer, Understand-It-Back quiz
Data posture	Accountless; private local storage; no document-content network calls
Safety position	Legal-information and comprehension aid; not professional legal advice

What makes it distinctive

- **From summary to comprehension.** It checks what the user understood rather than merely summarizing clauses.
- **From generic warnings to verifiable evidence.** Findings link to their original page, excerpt and review status.
- **From abstract clauses to bounded scenarios.** It models only consequences explicitly supported by verified contract terms.
- **From network dependence to private utility.** The critical workflow survives full airplane mode after installation.

Competitive honesty

Offline contract scanners and clause-highlighting applications already exist. The product should not claim world-first status. Its proposed differentiation is an end-to-end informed-understanding workflow, supported by demonstrable source traceability and genuine mobile inference.

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Document guide

This specification separates required hackathon behavior (H1) from later product extensions (P1+). Proposed performance targets and model selections must be validated on the intended demo device.

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01 / Product strategy & boundaries

Built for informed decisions under limited connectivity.

Problem statement

People routinely sign employment contracts, freelance agreements, installment plans, leases and service terms without a reliable way to identify financial obligations, restrictions, deadlines and unanswered questions. Cloud AI can help, but it introduces connectivity dependence, repeated processing costs and avoidable transmission of sensitive documents.

User promise

"Unawain bago pirmahan." Users should leave a session able to explain important written obligations in their own words, locate the original supporting language and list questions that remain unresolved.

Success outcomes

- Every major document-specific finding is traceable to a valid source passage.
- Users can correct OCR errors before calculations or explanations depend on those fields.
- A supported contract can be scanned, reviewed, understood and saved with the radio disabled.
- Users are never shown a deceptive safety, fairness or legality score.
- The application clearly abstains when a legal conclusion or scenario cannot be supported.

Out of scope for H1

NOT INCLUDED	RATIONALE
Electronic signature or contract execution	Would expand the product into identity, enforceability and workflow risks.
Lawyer substitution or legal certification	The model is not qualified to determine legality or guarantee enforceability.
Court outcome or rights prediction	Requires current legal authority and jurisdiction-specific expert validation.
Cloud AI, user accounts, automatic uploads	Contradicts the core offline/private hackathon demonstration.
Universal document coverage	Incompatible with reliable validation during a short build.

02 / Users, jobs to be done & use cases

PERSONA	HIGH-VALUE JOB	MATERIAL TO REVIEW
Freelance developer / designer	Understand what triggers final payment and how revisions or work ownership are defined.	Service agreements, statements of work
Entry-level employee	Find compensation, working conditions, notice provisions and terms to clarify.	Employment agreements
Installment buyer	Translate quoted payment terms into an understandable schedule.	Installment / financing agreements
Tenant	Find deposits, repair obligations and early move-out provisions.	Residential leases
Small business owner	Identify payment and cancellation conditions before accepting a supplier arrangement.	Supply / services contracts
Everyday document signer	Decode unfamiliar wording and formulate questions without sending files to a server.	General agreements

Priority end-to-end scenarios

- **UC-01 / Freelance payment.** Import a service contract, identify an acceptance-dependent payment clause, open its precise page, then produce clarification questions.
- **UC-02 / Installment obligations.** Scan payment terms, verify dates and amounts, explore an explicitly documented late-payment scenario, and view assumptions.
- **UC-03 / Employment wording.** Read a probation or termination clause and explain what the text says without independently declaring its legal validity.
- **UC-04 / Unreadable document.** Detect a blurry amount or missing page, ask for correction and block dependent conclusions.
- **UC-05 / Airplane mode.** Relaunch the installed app fully offline, analyze a fresh supported document and reopen the saved review.

> **H1 coverage decision**
> Fully exercise two narrow document families: freelance/service agreements and simple installment agreements. Employment and rental agreements may be imported for limited reading only; do not promise complete structured support.

03 / Scope, releases & requirements

Release strategy

RELEASE	COMMITMENT	EXIT CRITERION
H1 - Hackathon	Android local OCR + LLM, source-grounded terms, review, one supported scenario and understanding check, offline storage.	Fresh document demo in airplane mode.
P1 - Closed beta	More agreement types, ASK-document mode, review-report export, optional legal source packs.	Expert evaluation and measured language accuracy.
P2 - Public release	iOS, broader accessibility, large documents, version comparisons, refined privacy controls.	Device matrix testing and security review.
P3 - Partnerships	Opt-in organization deployments and curated legal-aid support resources.	Documented consent and partnership governance.

User stories - priority behavior

- **As a reader,** I can import pages without having to sign in or connect to internet.
- **As an uncertain reader,** I can correct extracted text before the app uses it.
- **As a cautious signer,** I can inspect the original evidence underlying any explanation.
- **As a learner,** I can answer questions about my own agreement and see the source for feedback.
- **As an installment buyer,** I can test only scenario rules that appear in the agreement.
- **As a privacy-conscious user,** I can securely delete local documents and their derivative analysis.

Priority legend

H1 must: required for the competitive live demonstration. **H1 stretch:** add only after the main offline pipeline is reliable. **P1:** explicitly deferred and not represented as completed.

04 / Functional requirements - complete register

ID	FUNCTION	ACCEPTANCE RULE	PHASE
F01	Welcome and privacy	Explains local processing, limitations and user control before first import.	H1
F02	AI readiness	Reports installed model, checksum, initialization status and offline readiness truthfully.	H1
F03	Camera capture	Multi-page capture with retake, rotate, crop and reorder.	H1
F04	File import	Reads supported local PDF and image files through system picker.	H1
F05	On-device OCR	Returns block text plus page and bounding location without remote service.	H1
F06	Text verification	User can correct unreadable money, dates, names and sections.	H1
F07	Document classification	Suggests supported type with manual override; may abstain.	H1
F08	Clause segmentation	Stores persistent page/section/block identifiers.	H1
F09	Structured extraction	Produces parties, amounts, obligations, deadlines and conditions using a strict schema.	H1
F10	Review findings	Surfaces important clauses, ambiguities and missing information.	H1
F11	Plain explanation	Explains verified facts without adding ungrounded legal assertions.	H1
F12	Original evidence	Tap any important finding to open exact original passage.	H1

ID	FUNCTION	ACCEPTANCE RULE	PHASE
F13	Consequence explorer	Runs bounded hypotheticals from confirmed contract parameters; shows assumptions.	H1
F14	Understanding check	Evaluates document-grounded quiz answers and cites source.	H1
F15	Question builder	Editable clarification questions tied to findings.	H1
F16	Local document vault	Store, reopen, rename, search and delete a review.	H1
F17	Inference diagnostics	Exposes local runtime/model and measured stage completion.	H1
F18	Ask-document chat	Conversational retrieval with citations and abstention.	P1
F19	Review export	Review, redact and explicitly export a report.	P1
F20	Revision comparison	Detect changes between distinct agreement versions.	P1
F21	Curated legal pack	Versioned verified legal reference material for supported topics.	P1
F22	Audio accessibility	Offline read-aloud and extended language support.	P1

05 / Evidence-first review specification

Make the original agreement the source of truth.

A document-specific output is permissible only when the source can support it. Every finding is a record with category, excerpt, page/block identifier, explanation, uncertainty state and suggested next question. The UI must distinguish observed text from model interpretation.

Required finding states

STATE	USER MEANING	BEHAVIOR
Needs attention	A materially important obligation or apparent contradiction is present.	Show evidence; do not call it illegal.
Needs clarification	A significant term is ambiguous, incomplete or insufficiently specified.	Identify the unanswered question.
Explained from source	Text was located and explained using a matched source passage.	Offer a jump to original clause.
Unable to verify	Source was unreadable, absent or the output lacked support.	Abstain and suggest recapture/review.

Worked example

Payment condition - fictional service agreement, page 3
SOURCE: "The final payment shall be released upon the Client's acceptance of all deliverables."

EXPLANATION: Final payment depends on the client accepting the deliverables.

UNCERTAINTY: This cited sentence does not say how long the client has to review them.

QUESTIONS: What are the objective acceptance criteria? Is there a review deadline?

Guardrails enforced by code

- Reject a generated source ID not present in the OCR database; match quoted text to recorded passages.
- Require extracted monetary values and dates to agree with readable source characters.
- If the user edits OCR, invalidate dependent facts, calculations, questions and cached conclusions.
- Do not infer overall contract safety, legal enforceability or whether the user should sign.
- Never convert absence of a clause into a claim that a legal right is absent.

06 / Signature learning experiences

A. Clause-to-consequence explorer

The user selects a verified clause and chooses a supported hypothetical. Numeric or date consequences are calculated in deterministic TypeScript using values reviewed against the agreement. The system must display the rule, inputs, source and assumptions.

FICTIONAL SCENARIO	WHAT THE APP MAY SHOW
Installment is PHP 2,000, due the 15th; document explicitly states PHP 100 fixed late charge.	On-time: PHP 2,000; late: PHP 2,100 for the shown conditions; reveal source of each figure.
Late-charge wording missing or OCR unreadable.	"Insufficient information"; do not invent any late fee.
Payment depends on client acceptance; no review deadline found in quoted clause.	Explain conditional trigger and ask for timeline clarification; do not assume payment will be withheld.

B. Understand-It-Back

Create short questions whose answers are supported by verified facts. The user can choose or describe an answer. Feedback states are Supported, Needs review and Cannot determine. Every correction has a "View source" action. This is a comprehension aid, not a formal exam.

C. Questions before signing

Generate concise, editable questions about gaps or conditions found in the source. Users may save or copy the questions to ask the other party or a qualified professional. Questions may include acceptance criteria, payment release, cancellation procedure, unclear fees or notice periods.

D. Source-linked navigation

Findings link directly to highlighted OCR blocks and the original PDF/image page. The document view supports zoom and visible text corrections. The UI should never make an AI paragraph appear to be contract text.

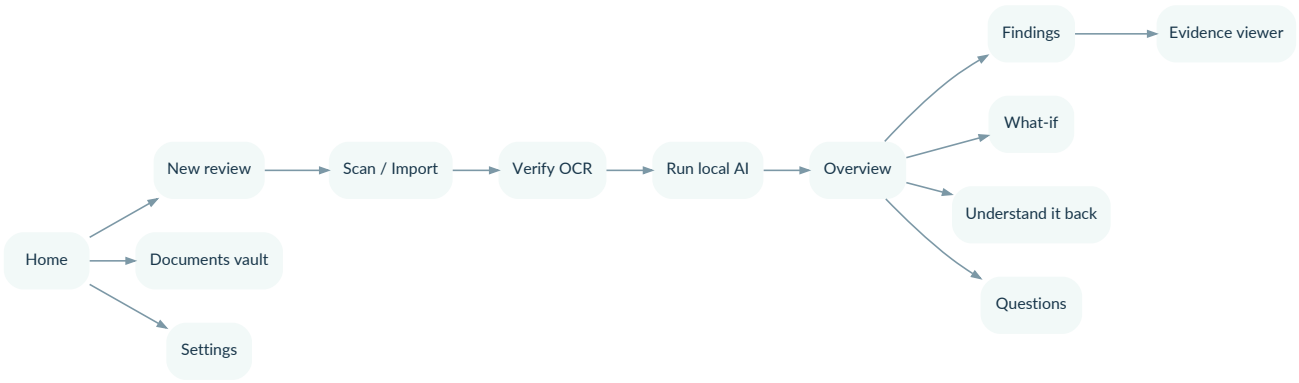
07 / Screen inventory & information architecture

SCREEN	TITLE	CORE CONTENT / ACTION	PHASE
S01	Launch	Warm startup; encrypted store / model state.	H1
S02	Welcome & privacy	Value, limitations, permissions and data commitment.	H1
S03	Model readiness	Model present, space/checksum, local inference check.	H1
S04	Home	New review, recent documents, offline status.	H1
S05	Add document	Camera, gallery and PDF entry.	H1
S06	Capture	Shoot, rotate, crop, retake, add page.	H1
S07	Preview	Page thumbnails; order and confirm.	H1
S08	OCR verification	Review/correct suspicious names, dates and amounts.	H1
S09	Analysis preferences	Document type, language, focus.	H1
S10	Processing	Stage status, cancel, safe recover.	H1
S11	Overview	Facts, findings, coverage, next actions.	H1
S12	Terms & obligations	Filterable extracted clauses.	H1
S13	Finding detail	Evidence, plain meaning, questions, uncertainty.	H1
S14	Original viewer	Page with highlighted evidence.	H1
S15	What-if explorer	Explicit scenarios and assumptions.	H1
S16	Understanding check	Quiz, feedback, source navigation.	H1

SCREEN	TITLE	CORE CONTENT / ACTION	PHASE
S17	Questions to ask	Editable question list with source links.	H1
S18	My Documents	Local search, reopen, rename, delete.	H1
S19	Settings	Language, model, storage, privacy, erase.	H1
S20	Ask document	Cited conversational questions.	P1
S21	Report review	Redact, preview, export.	P1
S22	Compare versions	Two-document clause differences.	P1
S23	Reference library	Curated offline legal materials.	P1
S24	Accessibility	Speech output and extended language preferences.	P1

08 / Navigation, wireframes & interaction flows

Information architecture



Core screen compositions (conceptual)

01 / HOME

My documents

- Review an agreement
- Recent reviews
- Privacy: on-device

Start a review

02 / OVERVIEW

Service agreement

- 6 terms identified
- 2 items to clarify
- Open original evidence

Review findings

03 / EVIDENCE

Payment terms

- Quoted clause - p. 3
- Plain-language meaning
- Question to ask

View source

Interaction requirements

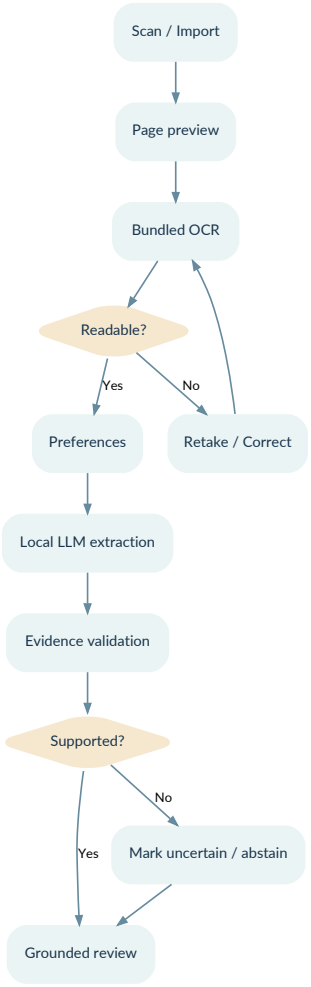
- Use plain English and Filipino; avoid legal jargon in primary UI labels.
- Link every finding card to exact evidence with a prominent View source action.
- Display processing state without implying that the AI has certified the agreement.
- Provide an editable correction path; use status color with text and icons, never color alone.

09 / Detailed user flows & state transitions

UF-01 / First use

Launch > privacy explanation > model installation/readiness > optional permissions > home. If required models are absent, do not claim full offline readiness.

UF-02 / Document-to-review



UF-03 / Inspect a finding

Overview > selected finding > summary + cited excerpt > open original page > optionally correct source > rerun affected findings > save question/review status.

UF-04 / Scenario and comprehension

Verified rule > scenario choice > verify parameters > local calculation > show assumptions/source; or verified fact > source-grounded question > user answer > supported / needs review / cannot determine > source jump.

UF-05 / Save and delete

Save encrypted analysis > reopen without new inference if hashes match > rename or review > explicit delete removes source, records and temporary derivatives.

10 / Technical stack & operating assumptions

Offline-first Android implementation, with documented compatibility checks.

LAYER	CHOICE	REASON / CAUTION
App framework	Expo + React Native + TypeScript	Use a custom development build for non-Expo native inference.
Routing & state	Expo Router + Zustand	Simple isolated flows; persist only required state.
Capture/import	expo-camera, document/image pickers	Camera/PDF input; PDF rendering via tested native library.
OCR	Bundled ML Kit Text Recognition v2 via Expo native bridge	Must be packaged on the device; inspect recognition bounds and script coverage.
LLM	llama.rn / llama.cpp with GGUF	Run a quantized small model on ARM64; use schema-constrained output if supported.
Model candidate	Qwen3 0.6B or 1.7B quantized, after device testing	Do not assume speed, memory or Filipino agreement accuracy.
Validation	Zod + TypeScript deterministic rules	Reject invalid source pointers, dates, amounts and result types.
Database	expo-sqlite with SQLCipher config	Encrypted database in custom native build; explicitly key at open.
File security	App-private encrypted file storage	SQLite encryption alone does not secure imported PDFs/images.
Keys	expo-secure-store / OS-backed secrets	Avoid hardcoded cryptographic keys.
Testing	Jest, RN Testing Library, Maestro + device profiling	Test inference, error paths and actual airplane-mode operation.

> **Critical build decision**
> Target one real Android ARM64 phone. Validate the precise Expo SDK, React Native New Architecture and llama.rn package versions before spending time on the complete UI. Bundled ML Kit and encrypted SQLite also require an appropriate native build.

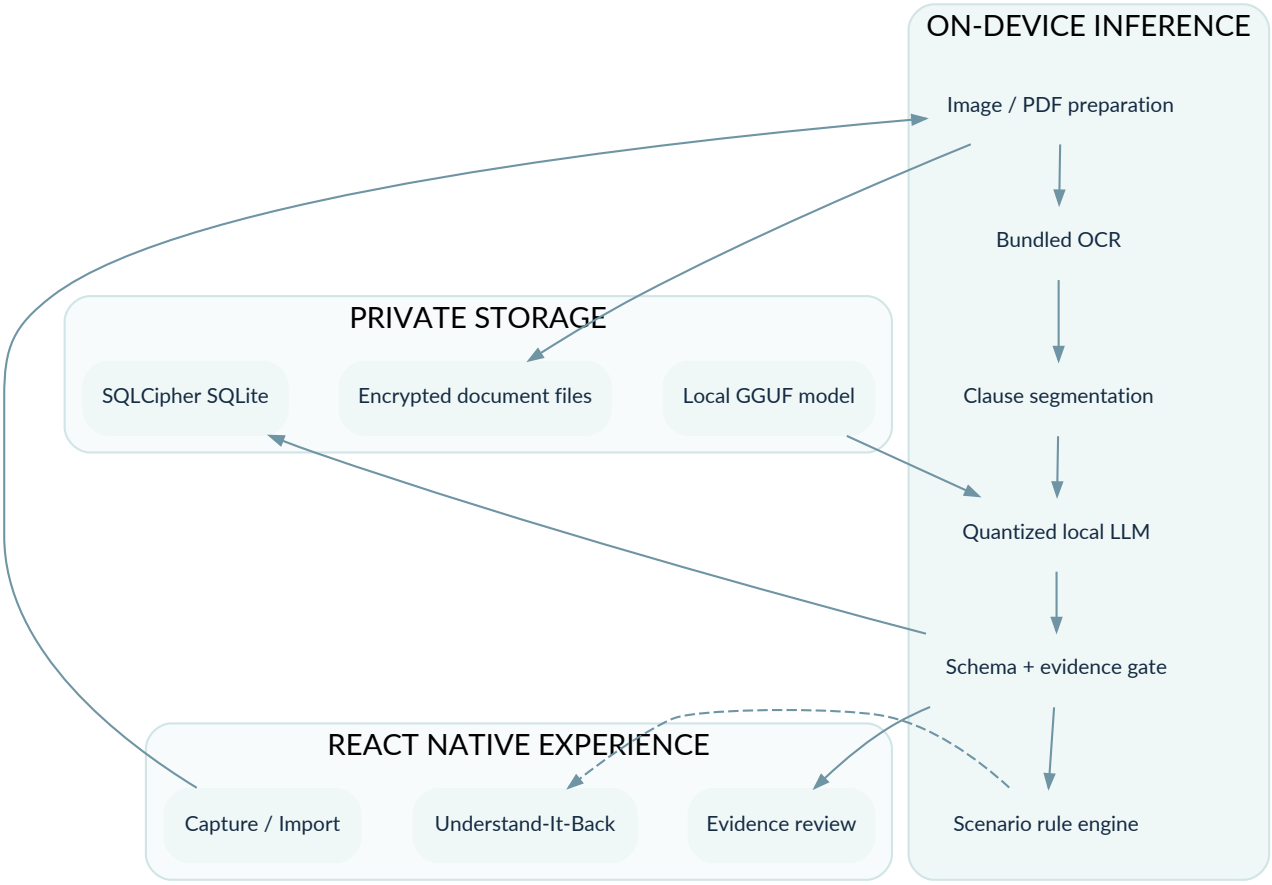
On-device quality strategy

Avoid prompting over an entire long contract at once. Perform extraction by page or clause, generate strict JSON, validate references, then aggregate document-wide verified facts. Use constrained prompts; provide an abstention outcome. Run financial mathematics outside the LLM.

Repository organization: keep Expo Router screens in app/; product features in src/features/; inference/validation in src/ai/; schema/storage in src/db/; native ML Kit bridge in modules/local-ocr/; and reproducible synthetic test fixtures in tests/.

11 / AI execution & system architecture

Offline data plane



Inference stages and determinism

STAGE	MECHANISM	EXPECTED RESULT
Acquire	Camera or file picker	Local page artifacts and format status
Recognize	Bundled OCR model	Text spans with page/location
Organize	Section parser + local LLM	Candidate facts/clauses
Validate	Schema + source matching	Accepted facts vs uncertain/abstained
Explain	Grounded local LLM	Plain language with citations
Simulate	Deterministic rules	Reproducible scenarios
Persist	Encrypted DB and files	Reopenable local session

12 / Inference contract, provenance & failure rules

Minimum structured finding

PROPERTY	EXAMPLE	RULE
type	payment_condition	Controlled category enumeration
normalized_value	upon_client_acceptance	Must not be invented
evidence.page	3	Existing document page
evidence.block_id	p3_b12	Existing OCR block identifier
evidence.quote	Exact excerpt	Must match recorded text span
explanation	Conditional payment	Grounded in cited text
review_status	needs_clarification	Limited allowed states
question	"How long is review?"	Explicitly framed for human clarification

Source provenance and cache keys

Each analysis run stores source file hash, OCR revision, source block IDs, model checksum/version, prompt version, and validation status. Subsequent OCR edits invalidate dependent findings. A cached result may be reused only when its full provenance key remains compatible.

Prompt-injection resilience

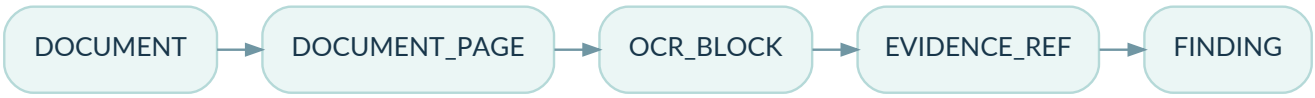
- All document text is untrusted task data, including statements that look like system instructions.
- Prompts delimit source blocks and ask for only constrained, evidence-linked output.
- The client rejects fabricated page IDs, malformed JSON and unsupported claims.
- Conclusions are not accepted merely because the model sounds confident.
- Retries are bounded; failed stages preserve the original document and show actionable errors.

Failure state matrix

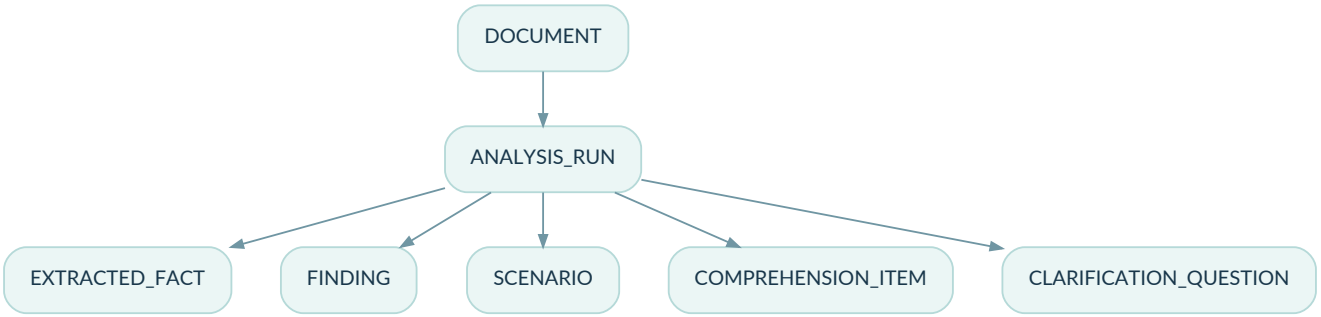
EVENT	PRODUCT RESPONSE
Blurred amount / poor OCR	Highlight region; request retake/correction; block dependent amounts.
Missing page or ambiguous source	Mark incomplete; do not declare a document fully analyzed.
Invalid LLM output	Retry once if safe; fall back to verified OCR and explicit uncertainty.
Low memory / unsupported model	Release model; report limitation; retain document safely.
Unsupported what-if rule	Show insufficient information and draft a question.
Network disabled	Continue all H1 core operations normally if readiness passed.

13 / Local data model & entity relationships

Provenance graph



Analysis records and dependents



All child relationships are one-to-many. An evidence reference connects a specific finding to a verified OCR block. User answers attach to comprehension items.

Persistence and integrity rules

- Encrypt document data and database at rest; use OS-backed keys and private app directories.
- Detect duplicates using local hashes; never upload hashes or source material automatically.
- Persist the original OCR text and user correction history for evidence auditability.
- Delete dependent facts, caches, images and temporary artifacts when the user erases a document.
- Keep source and analysis versions separate to avoid showing stale findings after a scan correction.

14 / Privacy, security & legal safety

Legal positioning
PirmaBago is a document comprehension and legal-information aid. It is not legal advice, a licensed attorney, a contract certification service, or an instruction to sign or refuse any document.

Privacy/security control register

CONTROL	REQUIRED IMPLEMENTATION
Local processing	No remote OCR, chat, inference or file upload is needed for H1.
User account	Not required; no cloud identity or background sync.
Encrypted storage	SQLCipher for records; authenticated encryption for source files.
Key management	OS keystore/keychain; avoid embedding app-wide secrets.
Permissions	Camera on capture; files via system picker; no broad file access.
Telemetry	Do not log OCR content, contracts, prompt bodies or PII.
Export	Human-reviewed, optional, explicit and redaction-aware (P1).
Deletion	Delete originals, derived records, temp files and relevant caches.
Model disclosure	Display model/runtime/license/version and state of offline readiness.
Retention	Local-only, user-directed; communicate device-loss implications.

Philippine privacy context

Republic Act No. 10173 (Data Privacy Act of 2012) and the National Privacy Commission implementing rules are relevant to any handling of personal data. On-device inference reduces transmission risk, but it does not itself prove legal compliance. Transparency, legitimate purpose, proportionality and appropriate security controls remain necessary.

Legal-information boundaries

- Describe what the agreement explicitly says; do not pronounce a clause legally valid or invalid.
- Use a curated, versioned legal source pack only when verified and legally reviewed (later release).
- For employment, tenancy, lending and consumer issues, statutory rights may exist outside the written clause.
- Flag uncertainty and suggest professional review for high-stakes, disputed or ambiguous issues.

15 / Localization, accessibility & UX principles

Language coverage

LANGUAGE / MODE	H1 STATUS	CAVEAT
English OCR and explanations	Primary	Evaluate across printed font sizes and layout types.
Filipino / Taglish explanations	Supported on tested fixtures	Natural language fluency is not proof of legal accuracy.
Filipino Latin-script OCR	Targeted tests	OCR handling may vary with print quality and names.
Bikol / other regional languages	Future exploration	Requires new datasets and evaluation before making claims.
Offline text to speech	Later	Must remain local and accessible when added.

Language style guide

PRODUCT CONCEPT	USER-FACING LABEL
Evidence	Tingnan ang nakasulat / View source
Ambiguous clause	Kailangan pang linawin / Needs clarification
What-if simulation	Ano ang mangyayari kung...? / What if...?
Understanding check	Naintindihan ko ba? / Check understanding
Unverifiable claim	Hindi matukoy sa dokumento / Cannot verify

Accessibility requirements

- Offer dynamic type support, screen-reader labels and sufficiently large touch targets.
- Communicate statuses using words and icons; never rely on red or green alone.
- Keep source quotes readable, searchable and zoomable.
- Use progressive disclosure: overview first, evidence and assumptions when requested.
- Never imply that an incorrect quiz answer means the user is unable to make decisions.

16 / Nonfunctional requirements & metrics

DIMENSION	TARGET / ACCEPTANCE TEST
Offline completion	100% of supported H1 critical journey after offline setup and cold restart.
Source traceability	Zero accepted material findings pointing to nonexistent pages or OCR blocks.
Deterministic math	All supported amount/date rules pass unit tests.
OCR correction	Critical money, date and party fields remain reviewable before simulation.
Latency	Engineering target: under 60 s for 2-4 clean pages on the selected device; measure, do not promise.
Reliability	No app crash or OOM on selected physical demo device for fixture set.
Responsiveness	Navigation and progress screen remain usable during native inference.
Persistence	Save/reopen works after app force-close and device restart.
Privacy	Zero document-content network requests during core review.
Transparency	Exact model, quantization and inference runtime disclosed.
Accessibility	Primary flows navigable with screen reader and high text scaling tests.

Measurable product metrics

- Grounded finding precision: share of displayed material findings supported by correct excerpts.
- Coverage recall: share of human-labeled important terms detected on benchmark fixtures.
- Correction burden: count of critical OCR edits per page.
- Abstention appropriateness: ability to withhold unsupported assertions.
- Comprehension usefulness: user can correctly identify an important obligation and locate its source.
- Technical health: warm/cold model load, inference duration, peak memory and crashes.

17 / Testing, evaluation & acceptance

Fixture set (recommended minimum)

FIXTURE	COUNT	WHAT IT TESTS
Fictional service agreements	10	Deliverables, IP, revisions, payment triggers
Fictional installment agreements	10	Fees, dates, schedules, calculation support
Clean PDFs and document images	10	Import, OCR, source mapping
Blurry / missing / incomplete pages	5	Correction and graceful abstention
English / Filipino mixed language	5	Language handling and ambiguity
Prompt-injection-like document text	5	Instruction separation and schema validation

Test suites

- Unit: monetary calculations, date offsets, hash checks, schema validators, source matching and cache invalidation.
- Integration: native OCR to LLM to evidence gate to database.
- UX: retake, OCR edits, source jump, uncertainty messages, deleting a document.
- Performance: peak RAM and latency over repeated sessions on the final Android device.
- Security: logs, plaintext leakage, temp files, and unintended document network requests.
- Offline: airplane mode, Wi-Fi off, force close, fresh document capture, inference and reopening.

Release acceptance gates

GATE	PASS CONDITION
G1 - Native inference	Quantized model executes meaningful extraction on physical device without internet.
G2 - Source truth	At least one unseen sample yields a correct, source-linked finding.
G3 - Safe limitation	One deliberately unsupported item produces an honest abstention.
G4 - Learning	One document-linked comprehension or consequence interaction works end to end.
G5 - Privacy/persistence	Document remains local and reopening works after app restart.
G6 - Demo	Repeatable full flow under airplane mode with disclosed model metadata.

18 / Implementation sequence & team plan

Four vertical milestones

MILESTONE	ENGINEERING WORK	DEFINITION OF DONE
01 / Prove local AI	Build Expo dev client, integrate llama.rn, load actual GGUF, run strict extraction offline.	Fresh local extraction on demo phone.
02 / Prove document evidence	Bundled OCR, PDF/camera source, stable block IDs, validator, overview and source viewer.	Tap finding to locate original clause.
03 / Prove differentiated UX	Scenario engine plus Understand-It-Back and clarification questions.	One safe interaction driven by verified source.
04 / Prove reliability	SQLCipher/files, safe failures, performance, cold-start offline rehearsal.	Complete demo without Wi-Fi or cellular.

Team allocation

ROLE	PRIMARY OWNER
Mobile/UX engineer	Capture, navigation, document viewer, review screens, user corrections
On-device AI engineer	OCR bridge, llama.rn integration, extraction, citations, inference diagnostics
Product & QA	Synthetic contracts, answer keys, legality constraints, demo, presentation

Build order and contingency

Validate runtime compatibility first. If memory or inference quality is insufficient, reduce model size, shorten document windows and constrain supported formats. If scenario modeling is unreliable, preserve the evidence review and prioritize a simple source-grounded understanding check. Never replace offline inference with a hidden API call just to pass the demo.

Implementation notes

- Create a minimal inference proof before styling all 19 H1 screens.
- Choose one device and pin actual package versions before the hackathon build.
- Use real inference on a previously unprocessed supported document in rehearsal.
- Keep test fixtures synthetic, with clearly documented expected answers.
- Disclose libraries, model names, licenses, training sources and any pre-hackathon preparatory work.

19 / Live demonstration & hackathon narrative

Suggested two-minute stage script

TIME	ACTION	WHAT JUDGES SHOULD SEE
00:00 - 00:20	Switch to airplane mode; cold-start app.	No network; installed model and readiness visible.
00:20 - 00:45	Photograph an unfamiliar printed service agreement.	Actual camera capture and local OCR progress.
00:45 - 01:10	Review a conditional-payment clause.	A grounded finding with exact page and quote.
01:10 - 01:40	Use Understand-It-Back.	Question and correction tied to the same source.
01:40 - 02:00	Save, reopen, disclose model.	Private persistent record and verifiable local inference.

Judging alignment

CRITERION	WEIGHT	PROOF
Problem & usefulness	25%	Important agreement; user gains plain-language comprehension.
Local AI	25%	Real OCR + LLM inference on device with radio off.
Technical execution	20%	Source mapping, corrections, persistence, graceful failure.
Innovation	15%	Verified understanding and bounded consequence exploration.
Demo and quality	15%	Simple UX, fresh source, understandable evidence.

Winning closing line

"When someone asks you to sign an important agreement, understanding it should not depend on expensive data, cloud access, or sending your private contract to an AI server."

Demo backup

Carry a prevalidated synthetic fallback contract and a second battery-charged Android device. A prepared input is acceptable as a contingency, but the inference results must be generated by the local model and not precomputed UI fixtures.

20 / Risks, mitigations & final definition of done

RISK	SEVERITY	MITIGATION
LLM invents explanation or legal authority	Critical	Evidence gate, controlled output schema, explicit abstention; no legal verdicts.
OCR corrupts amount/date	Critical	Highlight critical fields; manual verification; deterministic recalculation.
Model crashes on mobile	High	Benchmark early; smaller quantization; page-level processing; target one device.
Native Expo dependency conflict	High	Pin versions, custom development build, earliest integration test.
Sensitive plaintext leaks	Critical	Encrypted DB and files; temp hygiene; no content logs or telemetry.
Filipino legal-language weakness	High	Targeted evaluated fixtures; constrained scope; honest support labels.
Unclear novelty vs other scanners	Medium	Demonstrate understand-it-back and linked scenarios, not just summary.
Overextended feature scope	High	Lock H1 vertical slice; defer exports/chat/version compare.

Definition of done - H1

- Newly captured or imported supported document can be processed without internet.
- Substantive OCR and local LLM inference execute on the user device.
- Every displayed material finding has valid evidence; unreadable findings abstain.
- At least one safe what-if or comprehension flow is fully functional.
- The user can save, reopen and delete a protected local review.
- The final install survives airplane mode and cold-start testing.
- Models, libraries, version choices, limits and licenses are disclosed.

21 / References, assumptions & glossary

Primary technical and policy references

The links below support specific stack or regulatory context; they do not certify the application's legal accuracy, security or fitness for production use. Verify the versions selected during implementation.

Expo: Add custom native code
https://docs.expo.dev/workflow/customizing/

Expo: Development builds FAQ
https://docs.expo.dev/develop/development-builds/faq/

Expo SQLite and SQLCipher
https://docs.expo.dev/versions/v54.0.0/sdk/sqlite/

Google ML Kit model installation paths
https://developers.google.com/ml-kit/tips/installation-paths

Google ML Kit text recognition (Android)
https://developers.google.com/ml-kit/vision/text-recognition/v2/android

llama.rn repository and platform guidance
https://github.com/mybigday/llama.rn

National Privacy Commission: RA 10173
https://privacy.gov.ph/data-privacy-act-/

NPC: Implementing Rules and Regulations
https://privacy.gov.ph/implementing-rules-regulations-data-privacy-act-%202012/

Glossary

TERM	MEANING
Local AI	Meaningful inference occurs on the user device, without requiring a cloud AI API.
OCR	Optical character recognition, used to read text in images or PDF pages.
Evidence anchor	A reference connecting a finding to a source page/block/quote.
GGUF	A model-file format commonly used in llama.cpp deployments.
Quantization	Reduced-precision model storage/computation for deployment constraints.
Abstention	Refusal to draw an unsupported conclusion when evidence is inadequate.
H1 / P1	Hackathon release versus post-hackathon extension.

> **Product doctrine**
> Understand before you sign. Verify before you trust. Keep the document under the user's control.